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Oxygen-Driven Structural Reorganization by Trace Water Enabling Fast Li-Ion Transport in a Pliable Solid Electrolyte

Youngkyung Kim[1] | Jae-Seung Kim[2] | Kwang Ho Shin[1] | Jingyu Choi[1] | Beom Jin Park[1] | Jinyeong Choe[2] | Jiwon Seo[2] | Chanhyun Park[3] | Jeongjae Lee[4,5] | Dong-Hwa Seo[2] | Sung-Kyun Jung[1,6,7,8]

[1]Department of Materials Science and Engineering, Seoul National University (SNU), Seoul, Republic of Korea | [2]Department of Materials Science and Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea | [3]Institute of Physical Chemistry and Center for Materials Research (ZfM), Justus-Liebig-University Giessen, Giessen, Germany | [4]Department of Chemistry, University of Cambridge, Cambridge, United Kingdom | [5]Research Institute of Advanced Materials, Seoul National University, Seoul, Republic of Korea | [6]School of Transdisciplinary Innovations, Seoul National University (SNU), Seoul, Republic of Korea | [7]Research Institute of Advanced Materials (RIAM), Seoul National University (SNU), Seoul, Republic of Korea | [8]Institute for Rechargeable Battery Innovations Research, Seoul National University (SNU), Seoul, Republic of Korea

Correspondence: Dong-Hwa Seo (dseo@kaist.ac.kr) | Sung-Kyun Jung (naecard@snu.ac.kr)

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ABSTRACT

The hygroscopic property of solid electrolytes has long been regarded as a critical weakness, as exposure to ambient humidity typically induces decomposition and a loss of ionic conductivity. In this study, we reveal a counterintuitive yet constructive role of trace water in enhancing ion transport in a gallium-based pliable halide solid electrolyte. Controlled exposure of 2LiCl–GaF_3 to trace amounts of water introduces oxygen into the structure and reorganizes the local environments around gallium and lithium. This oxygen-driven structural reconfiguration forms Ga–O and Li–O linkages and diversifies the local environment of gallium by converting its coordination from sixfold to fourfold. It reduces the average Li–anion binding strength and flattens the Li-ion migration energy landscape, resulting in enhanced Li-ion transport. These changes collectively increase the ionic conductivity by more than two orders of magnitude, from 10^{-2} mS cm^{-1} to over 3 mS cm^{-1}, while imparting amorphous and deformable characteristics to the 2LiCl-GaF_3 electrolyte. This study demonstrates that the interaction between trace water and complex anion frameworks can be harnessed to simultaneously tailor structural and transport properties, offering a new design strategy for high-performance oxyhalide solid electrolytes.

1 | Introduction

With the growing demand for rechargeable batteries across applications ranging from small portable devices to electric vehicles, all-solid-state batteries (ASSBs) using solid electrolytes have emerged as leading candidates for next-generation batteries due to their potential for high energy density and safety [1–5]. Despite these advantages, one intrinsic property continues to hinder their practical use, which is the strong tendency of solid electrolytes to react with moisture. The absorption of water from the atmosphere often leads to structural decomposition and a severe loss of ionic conductivity, a universal challenge across sulfide, oxide, and halide systems [6–17]. In sulfide electrolytes, moisture generates toxic hydrogen sulfide gas and causes irreversible decomposition [6–8, 18]. In oxides such as the garnet-type $Li_7La_3Zr_2O_{12}$, air exposure forms lithium carbonate surface layers

Youngkyung Kima and Jae-Seung Kim contributed equally to this work.

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that raise interfacial resistance and hinder charge transfer [9, 10, 19, 20]. Halide electrolytes such as Li_3InCl_6 also form hydrated phases and suffer degradation in ionic conductivity [13, 14, 16, 21, 22]. To mitigate these issues, extensive research has focused on surface modifications [23–30], partial oxygen substitution [31–33], compositional tuning [34–36], and hybrid electrolyte systems [37, 38]. In addition, operation in ultra-dry environments with dew points below −40°C or even −60°C is often required, which inevitably raises production costs. Consequently, moisture sensitivity remains a critical bottleneck for the commercialization of solid-state batteries.

Recently, however, gallium-based complex anion materials, which are the inorganic-based pliable solid electrolytes [39–42], have shown exceptional behavior that challenges this conventional understanding. In 2LiCl–GaF_3, mild exposure to ambient moisture leads to an unexpected rise in ionic conductivity, reaching up to 12.9 mS cm^{-1}, rather than degradation [39]. A previous report speculated that this enhancement originates from proton conduction or lithium-ion conduction similar to that in aqueous solution systems [43–45]. Although such behavior suggests that the interaction between trace water and the halide framework may play a constructive rather than a destructive role, the fundamental origin of the conductivity increase and a systematic evaluation of the transference number have not been clarified yet.

Here, we verify that trace water serves as an effective oxygen source in the complex anion halide 2LiCl–GaF_3 solid electrolyte. Controlled exposure to moisture introduces oxygen into the structure, reorganizing the Ga-centered Ga–O–(F/Cl) network. Oxygen incorporation converts gallium coordination from octahedral to tetrahedral and diversifies the local lithium environments, mitigating the average strength of cation–anion interactions and flattening the energy landscape for lithium migration. Consequently, the ionic conductivity increases by more than two orders of magnitude, from 10^{-2} to 3 mS cm^{-1}, and the material becomes more amorphous and mechanically deformable.

These findings offer a new perspective on the hygroscopic behavior of solid electrolytes. Rather than being detrimental, trace water can act as a chemical mediator that introduces oxygen and tailors both structural and ion-transport characteristics. This study also provides fundamental insight into oxygen-mediated conduction in emerging oxyhalide electrolytes and establishes a new design principle for achieving high ionic conductivity through controlled interaction with moisture.

2 | Results and Discussion

2.1 | Enhanced Li-Ion Transport in 2LiCl–GaF_3 with Trace Water

To evaluate the influence of trace water on 2LiCl–GaF_3, we systematically controlled the contents of added water from 0.5 wt.% to 5 wt.% after synthesizing the fully dried sample. The electrolyte was prepared mechanochemically following published procedures, with an additional drying step at 120°C for both the precursors (LiCl and GaF_3) and the as-synthesized sample to remove residual moisture [39]. Under strictly anhydrous conditions, the obtained material exhibited a powder-like morphology (Figure 1a). The sample remained powdery up to 0.5 wt.% H_2O, whereas above 1 wt.% it developed a clay-like texture (Figure 1b; Figure S1). This observation implies that even small amounts of water are essential for achieving the pliable nature of 2LiCl–GaF_3 and that the softness observed in earlier studies likely originated from residual or adsorbed moisture in the precursors.

The ionic conductivity increased sharply with water addition, rising from 1.21×10^{-2} to 4.3 mS cm^{-1} at 25°C, accompanied by a reduction in activation energy from 0.609 to 0.253 eV (Figure 1c; Figure S2). To quantify the contribution of lithium ions to this enhanced conductivity, the Li^+ transference number (t^{Li+}) was measured using the Bruce–Vincent method (Figure 1d; Figure S3). For the powder samples (0 and 0.5 wt.% H_2O), LPSCl was used as a blocking layer (Li | LPSCl | electrolyte | LPSCl | Li). The interfacial stability of the LPSCl/electrolyte interface was verified, showing a stable response without continuous degradation (Figure S4). For samples with higher water contents (1 wt.%, 2 wt.%, 3 wt.%, 5 wt.% H_2O), which exhibit superior deformability, measurements were conducted in Li | garnet | electrolyte | garnet | Li symmetric-cell configuration. The t^{Li+} value decreased from 0.999 to 0.707 as the water content increased from 0 wt.% to 5 wt.%, suggesting partial participation of H^+ or other anionic species through limited salt dissociation similar to that in aqueous systems [43–45]. Despite this decrease, the effective lithium-ion conductivity ($\sigma_{Li,eff} = t^{Li+} \cdot \sigma_{total}$) continued to increase to 3.03 mS cm^{-1}, implying that water enhances Li-ion transport while possibly coexisting with minor protonic or anionic contributions. To examine the trend at higher water incorporation, 2LiCl–GaF_3 with a 10 wt.% H_2O sample was evaluated (Figure S5). In this case, t^{Li+} further decreased to 0.493, approaching values typical of liquid electrolytes, while the effective lithium-ion conductivity reached 3.81 mS cm^{-1}. The increase in $\sigma_{Li,eff}$ becomes significantly less pronounced, implying that the water-induced enhancement of Li^+ transport is saturated with excess water incorporation. Although further verification is required to isolate each contribution, these results collectively demonstrate that water plays a direct and active role in modifying Li-ion transport rather than simply serving as a source of conductive impurities such as protons.

Thermogravimetric analysis (TGA) provided additional insight into the chemical state of water in the material (Figure S6). The measured weight loss below 250°C was substantially smaller than the nominally added water amount (0.3 wt.% and 0.8 wt.% for samples with 3 wt.% and 5 wt.% water, respectively), indicating that water is not present as free molecular H_2O. Instead, it is likely incorporated as a coordinated or hydrogen-bonded species, consistent with the subsequent XPS, NMR, and DFT analyses discussed later.

The temperature dependence of ionic conductivity further revealed that water also affects the conduction mechanism. In the Arrhenius plots (Figure S7), anhydrous and low-water samples (≤0.5 wt.%) exhibited linear behavior, whereas samples containing ≥1 wt.% H_2O showed pronounced non-Arrhenius trends. This behavior resembles systems in which Li-ion transport is coupled to host lattice or segmental motions, such as polymer or glass electrolytes [46, 47]. To better describe this behavior,

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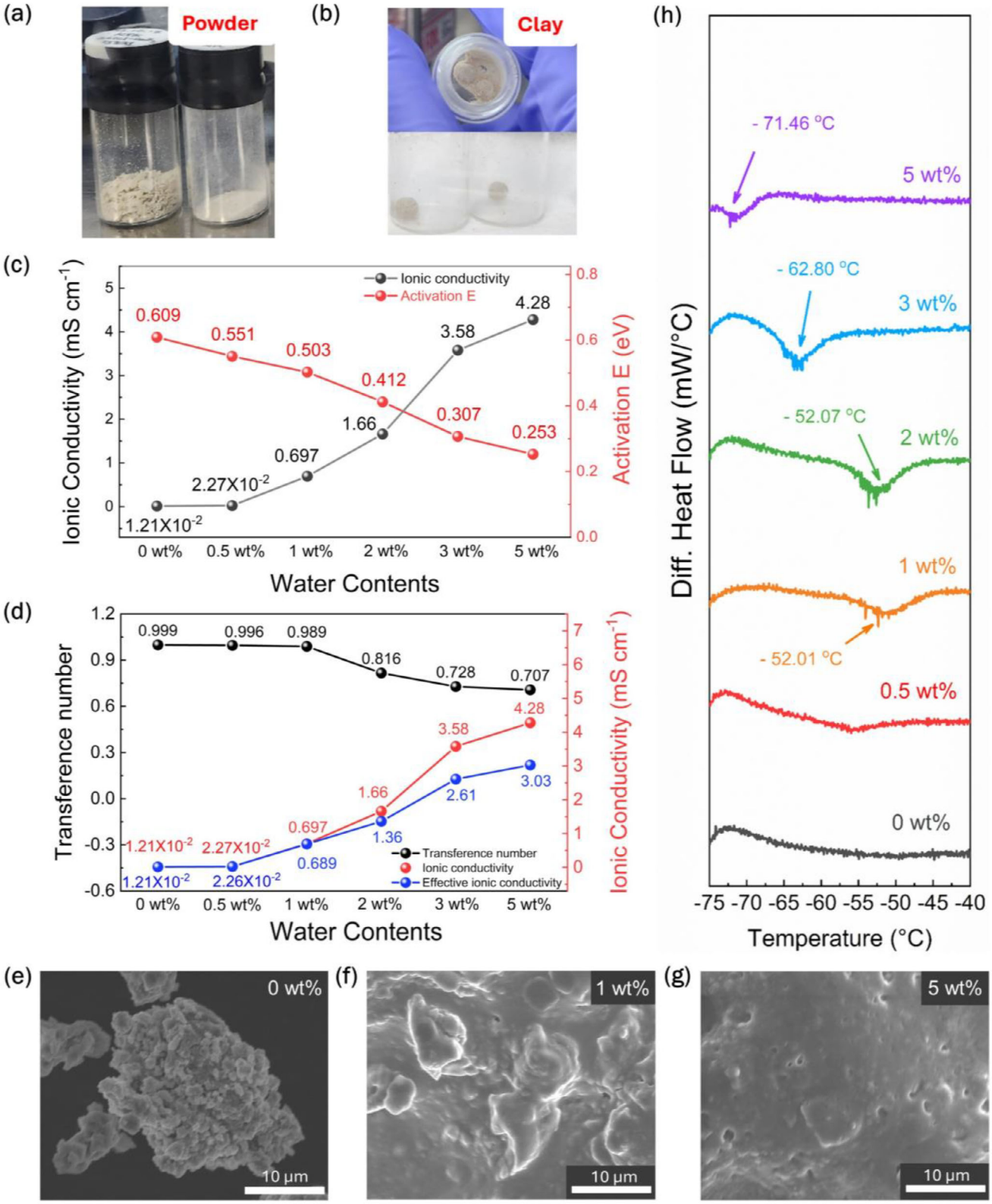

FIGURE 1 | Incorporation of trace water into the 2LiCl–GaF_3 electrolyte. Visual images of 2LiCl–GaF_3 (a) before (powder) and (b) after (clay) water incorporation. (c) Ionic conductivity of 2LiCl–GaF_3 depending on H_2O contents. (d) Li^+ transference number and effective ionic conductivity of water controlled 2LiCl–GaF_3. (e–g) SEM images of 2LiCl–GaF_3 (e) with 0 wt.%, (f) 1 wt.%, and (g) 5 wt.% H_2O. (h) Differential scanning calorimetry (DSC) profiles of water-controlled 2LiCl–GaF_3.

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the Vogel–Fulcher–Tammann (VFT) model [48, 49] was applied, which provided progressively better fits with increasing water content (Figures S8 and S9). Although the improved fit does not conclusively confirm polymer-like coupling, it suggests that trace water enhances the structural dynamics of the matrix, which in turn facilitates Li-ion migration.

Morphological and thermal analyses support this interpretation. Scanning electron microscopy (SEM) revealed enhanced particle agglomeration and interparticle connectivity with water addition (Figure 1e–g). Differential scanning calorimetry (DSC) measurements showed a systematic decrease in glass-transition temperature (T_g) with increasing water content (Figure 1h). Furthermore, the apparent density increases with water content (Table S1), suggesting improved packing and densification under the same pressure, consistent with water-induced viscoelastic deformation. These results indicate that water promotes amorphization and softening of the 2LiCl–GaF_3 network, leading to a more viscoelastic and deformable state. Together, these findings reveal that trace water simultaneously influences the microscopic ion-transport processes and the macroscopic mechanical properties of the electrolyte, and this coupling is examined in detail through structural analysis in the next section.

2.2 | Phase Evolution of 2LiCl–GaF_3 Upon Water Addition

Having established that water influences both ionic transport and mechanical response, we next investigated its effect on the structural evolution of 2LiCl–GaF_3 using X-ray diffraction (XRD) and complementary spectroscopies. Laboratory XRD revealed that the water-free (0 wt.%) sample was predominantly amorphous with minor residual LiCl peaks (Figure 2a), and these reflections weakened and nearly disappeared as the water content increased. Synchrotron-based diffraction identified additional crystalline phases, $LiGaCl_4$ (LGC) and Li_3GaF_6 (LGF) (Figure 2b), confirming that dehydrated 2LiCl–GaF_3 comprises a largely amorphous matrix over 60% with crystalline LGC, LGF, and LiCl (Figure 2c,d). Upon water addition, the intensities of both LiCl and LGC decreased significantly, as verified by Rietveld refinement (Figure 2d; Figure S10 and Tables S2–S7), indicating progressive amorphization [50, 51]. The relative phase fractions of LiCl and LGC decreased from approximately 11% and 6.6% to 0.1% and 0.5%, respectively. Correspondingly, Ga K-edge and Cl K-edge X-ray absorption near-edge spectroscopy (XANES) (Figure 2e; Figure S11) exhibited attenuated spectral features of LGC and LiCl, supporting the XRD observations. These results demonstrate that trace water promotes amorphization through weakened cation–halide coordination, a structural change that is expected to enhance Li-ion mobility.

To determine which phases, contribute to the observed conductivity enhancement, we independently examined the effect of water on each individual LiCl, LGC, and LGF phase. The ionic conductivity of LGF was intrinsically low (~10^{-7} S cm^{-1}) and further decreased upon water addition, without significant structural change (Figure S12). Similarly, the addition of 2 wt.% H_2O to LiCl reduced the conductivity and left the morphology unchanged (Figure S13). Thus, neither LGF nor LiCl can account for the water-induced increase in conductivity or the pliable character of 2LiCl–GaF_3.

In contrast, LGC exhibited distinct changes upon hydration. Increasing the H_2O content led to particle agglomeration (Figure S14) and a sharp increase in ionic conductivity, reaching the order of tens of mS cm^{-1} (Figure 2f; Figure S15). However, the Li^+ transference number (t^{Li+}) of hydrated LGC (Figure 2g; Figure S16 and Table S8) was very low (below 0.01 for 2 wt.% H_2O), implying that the increased conductivity mainly originates from proton conduction via the Grotthuss mechanism [45, 52] or from anion transport facilitated by solvation. Considering that solvation would typically enhance the mobility of both cations and anions, the observed negligible Li^+ transference number suggests that the conductivity is predominantly governed by the Grotthuss mechanism. This mechanism relies on proton hopping, which is significantly faster than the transport of other solvated ionic species [52, 53]. Despite the low t^{Li+}, the effective Li^+ conductivity increased from 10^{-3} to 0.4 mS cm^{-1}, suggesting partial enhancement of Li^+ mobility through localized hydration analogous to aqueous electrolytes [54, 55]:

$$Li^+ \ldots GaCl_4^- \xrightarrow{+H_2O} \left[Li(H_2O)_n\right]^+ \ldots GaCl_4^-$$

This indicates that introduced water molecules locally solvate Li^+, weakening the Li^+–$[GaCl_4]^-$ ion pairing and promoting partial dissociation, which results in a modest increase in Li^+ transport. XRD analysis of hydrated LGC (Figure 2h) showed pronounced peak broadening with increasing H_2O content (Table S9), confirming partial dissolution and structural disordering.

Although water-induced conductivity in LGC supports the idea that hydration can increase ionic transport, LGC accounts for only ~6.6 wt.% of 2LiCl–GaF_3, where the amorphous matrix dominates. Therefore, the overall conductivity enhancement in 2LiCl–GaF_3 cannot be explained solely by this crystalline phase. We infer that the substantial increase in conductivity and mechanical softening primarily arises from interactions between H_2O and the amorphous framework, which provides a flexible Ga–O–(F/Cl) network capable of accommodating water-induced coordination changes without crystallographic constraint. This structural adaptability and its connection to local coordination evolution are examined in detail in the next section.

2.3 | Local Structure Reorganization in 2LiCl–GaF_3 and $LiGaCl_4$

To clarify the distinct role of water in 2LiCl–GaF_3 and LGC, we performed solid-state nuclear magnetic resonance (ssNMR) spectroscopy to probe their local structural environments. The ^{71}Ga NMR spectrum of LGC (Figure 3a) exhibits a signal corresponding to tetrahedrally coordinated Ga (~245 ppm), and this fourfold coordination remains essentially unchanged upon water addition (Figure S17) [39]. Consistently, Ga K-edge X-ray absorption spectroscopy (XAS) shows no noticeable variation in the first coordination shell around Ga (Figure S18), indicating that the local environment of Ga in LGC is largely insensitive to water incorporation.

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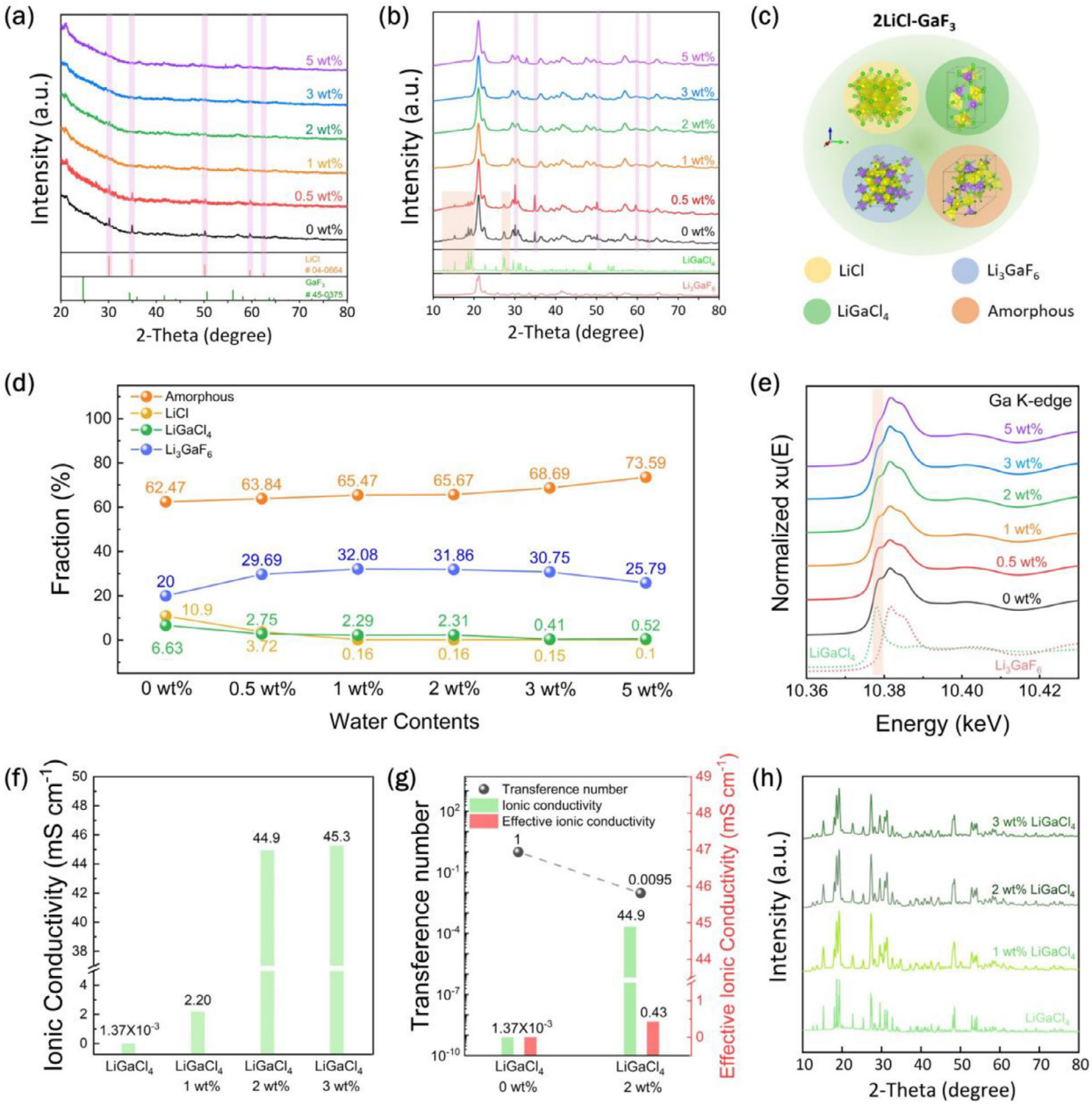

FIGURE 2 | Phase evolution of 2LiCl–GaF_3 with the addition of water. (a) Laboratory X-ray diffraction (XRD) patterns and (b) synchrotron XRD patterns of water-controlled 2LiCl–GaF_3 with reference Bragg positions for LiCl (purple) and $LiGaCl_4$ (orange). (c) Scheme of the 2LiCl–GaF_3 composition. (d) Phase fractions of each phase in 2LiCl–GaF_3. (e) Ga K-edge XANES spectra of water-controlled 2LiCl–GaF_3. (f) Ionic conductivity of water-controlled $LiGaCl_4$ (with 0 wt.%, 1 wt.%, 2 wt.%, 3 wt.% H_2O). (g) Li^+ transference number, ionic conductivity, and effective ionic conductivity of $LiGaCl_4$ before (0 wt.%) and after (2 wt.%) water addition. (h) Synchrotron XRD patterns of water-controlled $LiGaCl_4$.

In contrast, the ^{7}Li NMR spectra of LGC show two distinct environments, a broad resonance from less mobile Li^+ species (−1.2 ppm) and a sharp peak from highly mobile Li^+ ions (−0.5 ppm) (Figure 3b). The broad peak becomes more intense and progressively narrows with increasing water content, reflecting both a higher fraction and greater mobility of fast Li^+, consistent with the rise in effective Li-ion conductivity (Figure 2g). To further examine the role of water, deuterium-labeled water (D_2O) was introduced, and ^{2}H NMR reveals a sharp signal near 4.7 ppm that grows with water content, indicating the presence of rapidly diffusing free D_2O molecules that form hydration structures with Li^+ in LGC (Figure 3c) [56, 57]. These observations confirm that the enhanced total conductivity of hydrated LGC primarily arises from proton transport, while the contribution of Li^+ transport remains minor due to the low transference number ($t^{Li+} < 0.01$) (Figure 2g). The particle agglomeration observed in hydrated LGC can be rationalized by analogy with natural clay minerals, where partial dissolution and the formation of hydrogen-bonded networks impart cohesion and viscoelasticity [58–60].

Meanwhile, the NMR results of 2LiCl–GaF_3 show a fundamentally different trend. The ^{71}Ga NMR spectrum of the anhydrous sample (0 wt.%) displays two Ga coordination states, a dominant octahedral signal (~ −68 ppm) and a weaker tetrahedral

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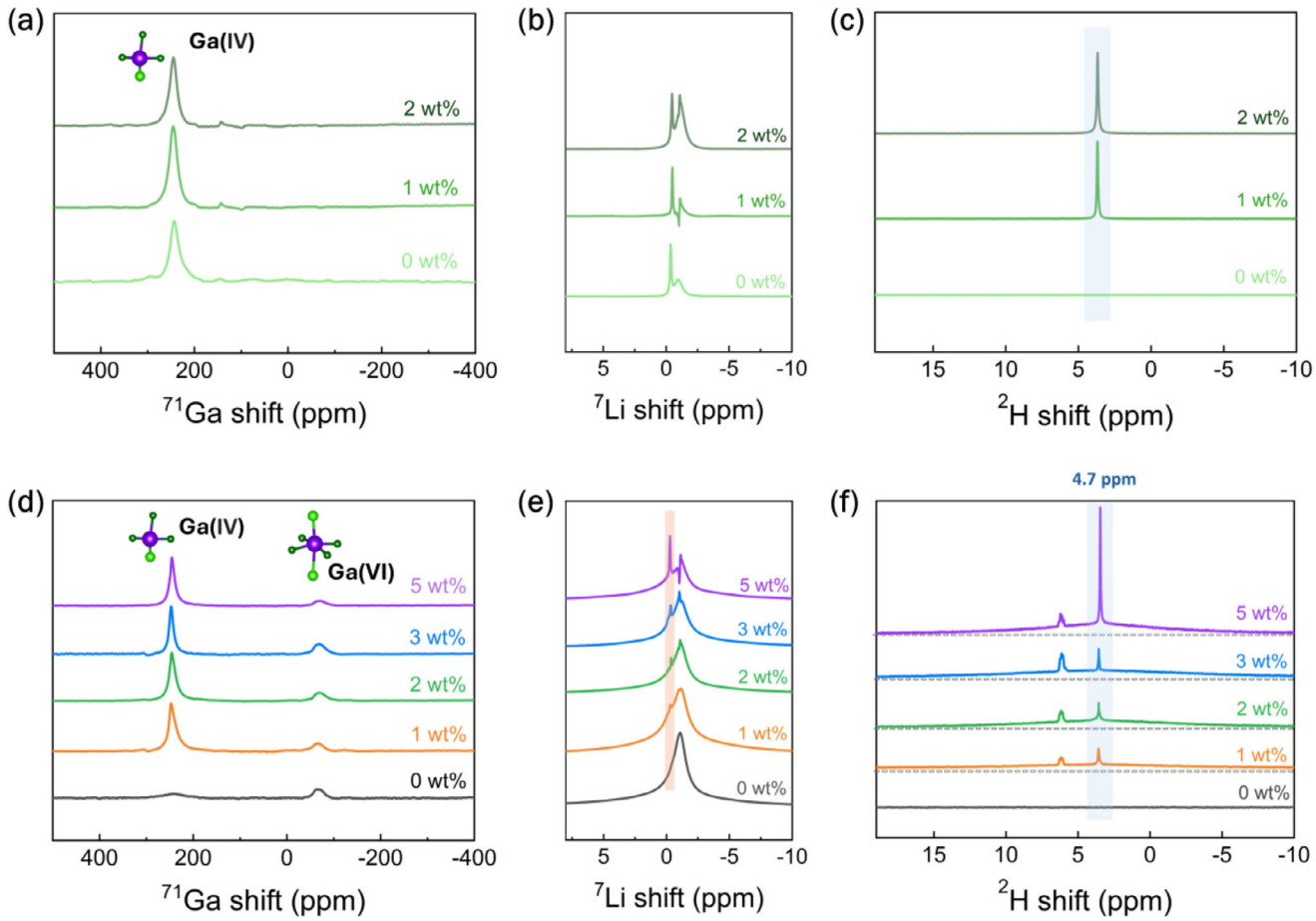

FIGURE 3 | Reconfiguration of local coordination around Ga and Li in 2LiCl–GaF_3 and $LiGaCl_4$ with addition of water. (a–c) NMR spectra of water-controlled $LiGaCl_4$ (with 0 wt.%, 1 wt.%, and 2 wt.% H_2O) for (a) ^{71}Ga, (b) ^{7}Li, and (c) ^{2}H nuclei. (d–f) NMR spectra of water-controlled 2LiCl–GaF_3 (with 0 wt.%, 1 wt.%, 2 wt.%, 3 wt.%, and 5 wt.% H_2O) for (d) ^{71}Ga, (e) ^{7}Li, and (f) ^{2}H nuclei.

component (~245 ppm) (Figure 3d), indicating that both Li_3GaF_6 (intrinsically sixfold Ga coordination, Figure S17) and the amorphous matrix are primarily composed of sixfold-coordinated Ga species under dry conditions. As the water content increases, the fraction of tetrahedral Ga gradually rises (Figure 3d; Table S10). As discussed later, the increase in tetrahedral coordination likely reflects the formation of more bulk bonding environments with increasing water content. This change is further supported by extended X-ray absorption fine structure (EXAFS) analysis (Figures S19 and S20), which shows broadening and intensity reduction of the Ga first-shell, confirming diversification of the local Ga coordination environment. O K-edge XAS further supports the formation of oxygen-containing species upon water addition (Figure S21). For 0 wt.%, only a negligible oxygen signal is observed, which we attribute to trace water unavoidably absorbed during sample loading into the vacuum chamber. In contrast, 3 wt.% exhibits pronounced O K-edge features, indicating the formation of oxygen-containing species by water incorporation. Specifically, the peak at 535 eV is consistent with the characteristic signal of liquid water [61], suggesting that a free-water component associated with solvation is present. On the other hand, the sharp peak at 532.5 eV cannot be explained by liquid H_2O alone. Considering the Ga K-edge EXAFS (Figures S19 and S20) evidence for local structural evolution, we attribute this feature to oxygen species more strongly coordinated to Ga and/or Li cations. Notably, the overall spectral shape is clearly distinct from that of Ga_2O_3, suggesting that the oxygen is not present as oxide (O^{2-}) but is more likely incorporated in hydroxylated or hydrated forms such as —OH or —OH_2.

The ^{7}Li NMR spectra show that the sharp, fast-ion component increases with water addition (Figure 3e), indicating an expanding population of highly mobile Li^+ species. Together, these findings suggest that water incorporation reorganizes the local coordination of both Ga and Li, thereby promoting Li-ion transport.

Deuterium NMR provides further insight into the state of incorporated water (Figure 3f). Three distinct ^{2}H environments are identified as a sharp resonance at 4.7 ppm corresponding to freely mobile water, consistent with the O K-edge XAS results (Figure S21), a broader signal at a similar chemical shift assigned to surface-bound or matrix-confined water with reduced mobility, and an additional split peak near 6 ppm consistent with H···F coupling. The latter feature indirectly implies the presence of hydroxylated species such as —OH or —OH_2 groups bound to cationic sites as residual species of H–F coupling [62, 63]. The formation of these oxygen-containing coordination structures is also verified by X-ray photoelectron spectroscopy (XPS), which reveals Ga—O/Ga—OH and Li—O/Li—OH bonding (Figure S22),

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and by time-of-flight secondary ion mass spectrometry (TOF-SIMS), which detects GaO^-, $GaOH^-$, $GaOH_2^-$ species (Figure S23).

To further verify the potential existence of hydroxylated oxygen species, NMR was performed on the 2 wt.% sample after re-drying at 120°C (Figure S24). The ^{71}Ga NMR spectra show that the fraction of tetrahedrally coordinated Ga remains higher than in the anhydrous (0 wt.%) sample even after drying, and the ^{7}Li NMR spectra still indicate the presence of a fast Li^+ component. In addition, the post-dried ^{2}H NMR spectrum reveals residual bound signals, demonstrating that not all water-derived species are removed as free H_2O. The persistence of altered Ga and Li local environments after drying provides indirect evidence that oxygen is retained in strongly coordinated, hydroxylated or hydrated forms (—OH or —OH_2), rather than existing solely as mobile free water.

Importantly, the evolution of these signals clarifies how water is incorporated within the material. The fraction of tetrahedral Ga increases steadily up to 3 wt.% H_2O, while the signal corresponding to free water near 4.7 ppm remains negligible. This indicates that at low water concentrations, H_2O primarily reacts within the amorphous matrix, coordinating to Ga and Li as hydroxylated species that drive local structural reorganization. Beyond 3 wt.%, however, the pronounced growth of the free-water peak suggests that excess water leads to partial dissolution and the formation of molecular H_2O.

These results collectively demonstrate that trace water modifies the local coordination environment through controlled hydroxylation rather than simple hydration. The resulting structural diversification is believed to flatten the energy landscape for Li^+ migration and enhance ionic conductivity. This water-mediated reorganization, in which trace water acts as an oxygen donor that reconstructs the framework and stabilizes flexible, fast ion-conducting networks, is reminiscent of oxygen incorporation strategies recently reported for oxyhalide electrolytes.

2.4 | Calculated Structural Changes and Li Transport in Water-incorporated 2LiCl–GaF_3

Guided by the experimental observations, amorphous structures of 2LiCl–GaF_3 and water-incorporated 2LiCl–GaF_3 were generated using melt–quench ab initio molecular dynamics (AIMD) simulations to examine the relationship between local structure and ionic transport. To minimize configuration bias, six independent amorphous models were constructed based on $LiGaCl_4$ and Li_3GaF_6 frameworks (three each). The detailed structure-generation protocol and statistical sampling procedures are described in Figures S25 and S26 and in the Section 4.4.

Representative atomic configurations for 2LiCl–GaF_3, 2LiCl–GaF_3 (+ 2 wt.% H_2O), and 2LiCl–GaF_3 (+ 4 wt.% H_2O) are presented in Figure 4a. In the water-incorporated models, oxygen atoms coordinate to Ga centers either as molecular H_2O or as hydroxyl (OH^-) species, forming stable Ga–O motifs within the amorphous network. Statistical analyses of all six models of each composition reveal that increasing water content leads to a marked reduction in Ga–F coordination and a modest decrease in Ga–Cl coordination, accompanied by a progressive rise in Ga–O coordination (Figure 4b). Also, continuous symmetry measure (CSM) analysis [64] further confirms that these coordination changes are accompanied by a transition from octahedral to tetrahedral Ga environments upon water incorporation (Figure S27). These trends suggest that oxygen preferentially interacts with Ga sites formerly coordinated by F^-, leading to partial replacement within the local environment rather than complete substitution of halides. This interaction results in a more diverse Ga-centered coordination network that mirrors the experimental observation of amorphization.

Such oxygen incorporation not only diversifies the Ga coordination network but also alters the electrostatic landscape experienced by Li^+, as reflected in the enhanced Li^+ diffusivity. To quantify the impact of this structural reorganization on ion mobility, Li^+ dynamics were analyzed using AIMD simulations. Probability-density maps show that the spatial distribution of Li^+ broadens with increasing water content (4 wt.% > 2 wt.% > 0 wt.%), signifying enhanced dynamic migration (Figure 4c). Consistent with this, the mean square displacement (MSD) shown in Figure 4d confirms a steady increase in Li^+ diffusivity at 600 K from 4.58×10^{-6} cm^2 s^{-1} in anhydrous 2LiCl–GaF_3 to 8.07×10^{-6} and 9.86×10^{-6} cm^2 s^{-1} for the 2 wt.% and 4 wt.% H_2O-incorporated models, respectively. The ensemble-averaged MSD results further substantiate that oxygen incorporation consistently promotes Li^+ mobility across different configurations (Figure S28). To quantify the strength and distribution of Li–anion bonding, the integrated crystal orbital Hamilton population (ICOHP) analysis shows that oxygen incorporation weakens the average Li–anion interaction while increasing Li-site energy dispersion, collectively promoting Li-ion migration (Figure S29).

To understand the anionic coordination environments governing Li^+ transport, radial distribution function (RDF) analyses were performed over 200 ps trajectories (Figure 4e). Upon water incorporation, a distinct Li–O correlation emerges, in addition to the existing Li–F and Li–Cl interactions. The average Li–anion distances follow the order F < O < Cl, implying that oxygen introduces an intermediate-strength electrostatic interaction between Li^+ and the surrounding anions. Integration of $r^2g(r)$ yields the coordination numbers and relative fractions (Figure 4f, please note the Section 4.4). Because coordination is determined by r^2-weighted integration rather than peak height, subtle redistribution of anion correlations toward larger distances can alter the relative coordination fractions (Figure S30 and Table S11). The results show that Li–O coordination contributes more than 30% of the total Li–anion coordination, revealing substantial anionic diversification within the conduction environment. Notably, although oxygen constitutes less than 10% of the Ga-centered coordination, it exerts a disproportionately large influence on Li^+ transport, highlighting the crucial role of localized oxygen coordination in modulating ionic conduction.

Therefore, the enhanced Li^+ diffusivity observed upon water incorporation can be rationalized by the synergistic effects of anion interaction and structural disorder, rather than attributed solely to compositional changes. Oxygen, whose electronegativity lies between that of Cl^- and F^-, modifies the local electrostatic potential by partially replacing monovalent halide anions with divalent O^{2-} species. This substitution diversifies Li^+

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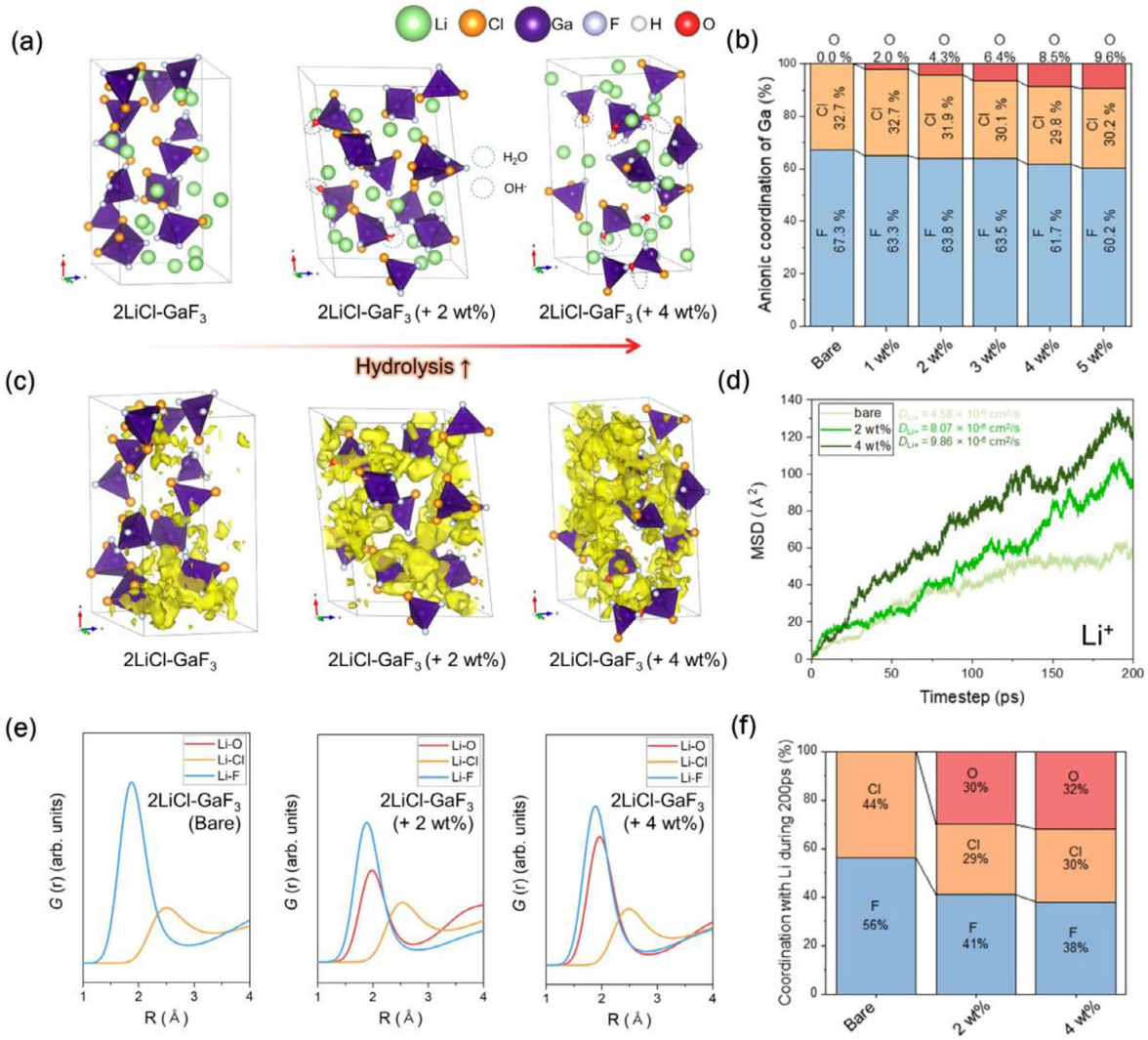

FIGURE 4 | Calculated local coordination evolution and Li^+ transport acceleration in 2LiCl–GaF_3 by oxygen-incorporation. (a) Calculated structure of water-controlled 2LiCl–GaF_3 with bare (left), 2 wt.% (middle), 4 wt.% (right). (b) Anionic coordination ratios of Ga with F^-, Cl^-, and O^{2-} in water-controlled 2LiCl–GaF_3 structures. (c) Li^+ probability density ($P = 0.2P_{max}$) water-controlled 2LiCl–GaF_3 with bare (left), 2 wt.% (middle), 4 wt.% (right). (d) Mean square displacement (MSD) of Li^+ in AIMD simulation (600 K, during 200 ps) in water-controlled 2LiCl–GaF_3 structures. (e) Radial distribution function (RDF) profile in AIMD simulation (600 K, during 200 ps) between Li and O/Cl/F^- in water controlled 2LiCl–GaF_3 with bare (left), 2 wt.% (middle), 4 wt.% (right) structures. (f) Coordination between Li and F^-, Cl^-, and O^{2-} in water-controlled 2LiCl–GaF_3 structures during 200 ps AIMD simulations. Coordination fractions are determined by integrating $4\pi\rho r^2 g(r)$ up to the first minimum.

coordination environments and weakens the average Li–anion binding strength, thereby facilitating easier hopping between adjacent sites. Consequently, the enhanced Li^+ transport arises from the combined effects of anion substitution and structural reorganization, which flatten the potential-energy landscape and create continuous Li^+ migration pathways throughout the amorphous matrix.

2.5 | Electrochemical Performance of Water-Incorporated 2LiCl–GaF_3

To investigate how water incorporation affects the electrochemical properties of 2LiCl-GaF_3, the oxidation and reduction stability were first evaluated as a function of water content. An electrolyte composite was prepared by mixing the water-controlled 2LiCl–GaF_3 electrolyte with 1.5 wt.% carbon nanofibers (CNF), and cyclic voltammetry was performed using the cell configuration electrolyte composite | Li_3InCl_6 (LIC) | LPSCl | In–Li (Figure 5a; Figure S31). Under the present conditions, where water exists only in trace amounts, the electrochemical stability window remained comparable to that of the 0 wt.% sample. However, it is expected that excessive water introduction would eventually narrow the stability window as the system approaches the behavior of aqueous electrolytes. In addition, changes in local acidity/basicity associated with water incorporation may also influence the stability window.

Rate capability was compared using representative samples with 0 wt.%, 2 wt.%, and 5 wt.% H_2O (Figure 5b–d). Because of the intrinsically low ionic conductivity of the 0 wt.% sample (1.2×10^{-2} mS cm^{-1}), which restricts assessment at room

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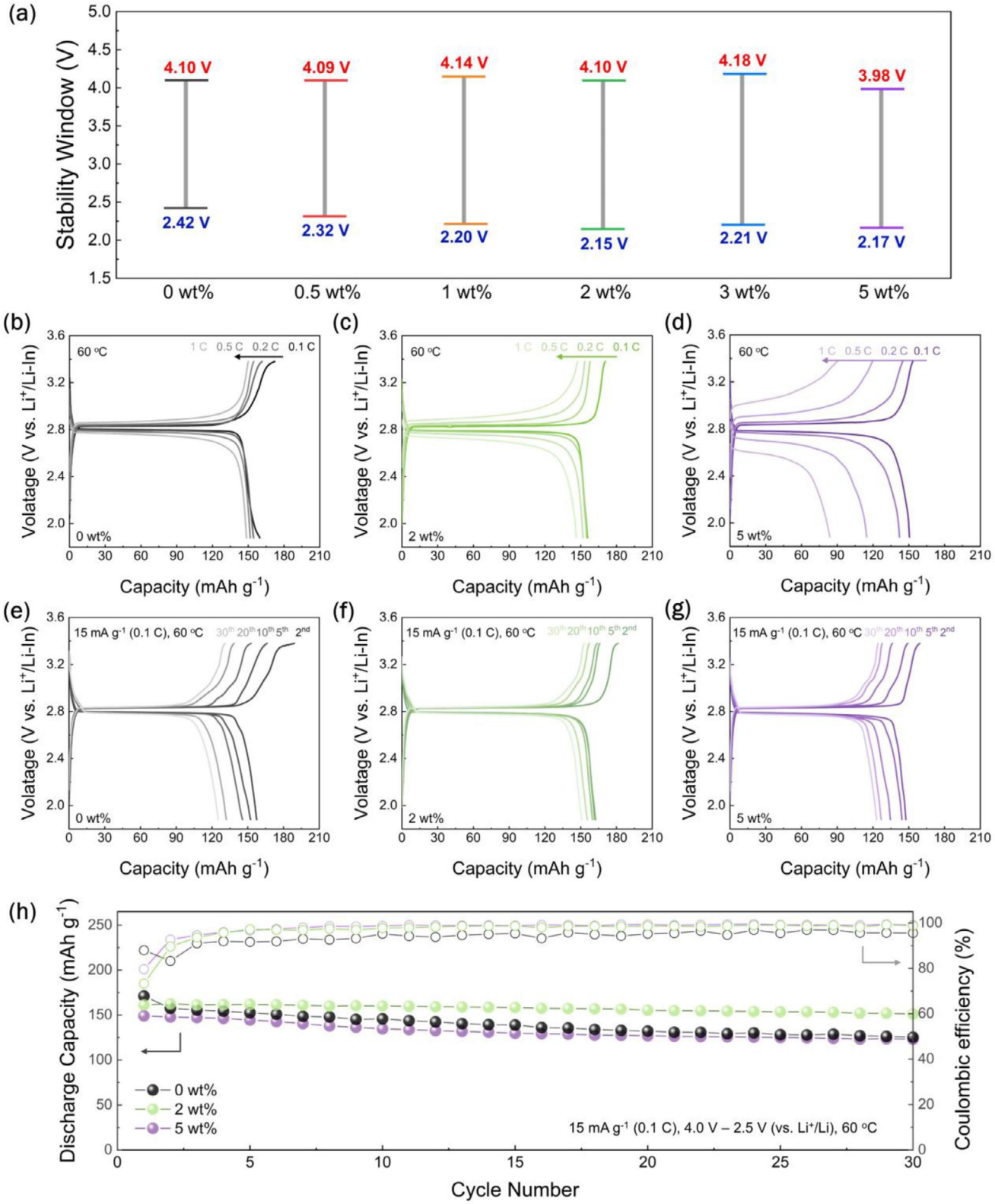

FIGURE 5 | Electrochemical test of 2LiCl-GaF_3. (a) Electrochemical stability window of water added 2LiCl–GaF_3. (b–d) Rate capability curves for 2LiCl-GaF_3 with 0 wt.% (b), 2 wt.% (c), and 5 wt.% (d) H_2O at C-rates of 0.1C, 0.2C, 0.5C, and 1C (1C = 150 mA g^{-1}) at 60°C. (e–g) Electrochemical profile of (e) 0 wt.%, (f) 2 wt.%, (g) 5 wt.% 2LiCl–GaF_3 at 60°C. (h) Cycling performance of 0 wt.%, 2 wt.%, and 5 wt.% at 15 mA g^{-1} (0.1C) and 60°C.

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temperature, the measurements were conducted at 60°C at 0.1C, 0.2C, 0.5C, and 1C. Although ionic conductivity increased with higher water content, the performance at high C-rates deteriorated as the water content increased. A similar trend was also observed during 2C cycling at 60°C (Figure S32). The 5 wt.% sample, which is expected to undergo severe side reactions and interfacial degradation due to its high fraction of free water (Figure 3f), exhibited the poorest capacity retention. Excluding this sample, the 2 wt.% sample delivered a higher discharge capacity, consistent with its higher ionic conductivity. However, with continued cycling, the 0 wt.% sample exhibited better capacity retention. This is likely associated with water-enabled side reactions within the electrolyte and at the interfaces, which can be further accelerated at high rates due to increased polarization and elevated local overpotentials.

In contrast, a distinct trend was observed during cycling at a lower current density of 0.1C (15 mA g^{-1}). As shown in Figure 5e–h, the 2 wt.% sample exhibited the most stable cycling behavior with the smallest capacity decay, whereas the 5 wt.% sample showed the lowest capacity retention. These results suggest that trace water can be beneficial in low current cycling by enhancing both ionic conductivity and mechanical softness, which improves interfacial contact compared to the 0 wt.% sample. For the 5 wt.% sample, however, where ^2H NMR (Figure 3f) indicates a large fraction of free water, water-induced side reactions and interfacial degradation are likely more pronounced. In particular, two interfacial configurations in contact with 2LiCl–GaF_3 were constructed and compared (LFP cathode/2LiCl–GaF_3 interface and Li_3InCl_6/2LiCl–GaF_3 interface). Water-related reactivity was more pronounced at the Li_3InCl_6 interface than at the LFP interface (Figure S34). The EIS results showed a more pronounced time-dependent increase in resistance for 2LiCl–GaF_3 containing 5 wt.% H_2O than for anhydrous 2LiCl–GaF_3, indicating that the presence of water facilitates parasitic reactions with Li_3InCl_6 and at its interface. Accordingly, the poorer cycling performance under high water conditions (at 5 wt.%) can be attributed to free water-induced interfacial reactions with Li_3InCl_6, which is known to suffer conductivity loss and form reaction byproducts upon exposure to moisture [13, 65], thereby accelerating interfacial resistance growth and increasing polarization. Therefore, adding an appropriate amount of trace water to supply oxygen to the local structure is crucial to ensure interfacial stability while maintaining high ionic conductivity and mechanical flexibility.

3 | Conclusion

Contrary to the long-held belief that moisture degrades solid electrolytes, this study reveals that trace water can instead serve as a constructive mediator that reorganizes the structure and activates ionic transport in 2LiCl–GaF_3. As illustrated in Figure 6, two interrelated pathways govern this water-driven behavior. The first is a localized solvation process occurring throughout 2LiCl–GaF_3, in which minor crystalline domains such as $LiGaCl_4$ partially dissolve to form gel-like regions that facilitate proton conduction via the Grotthuss mechanism. The second, and dominant, pathway occurs in the amorphous matrix, where trace water acts as an oxygen donor to form hydroxyl (—OH/—OH_2) species

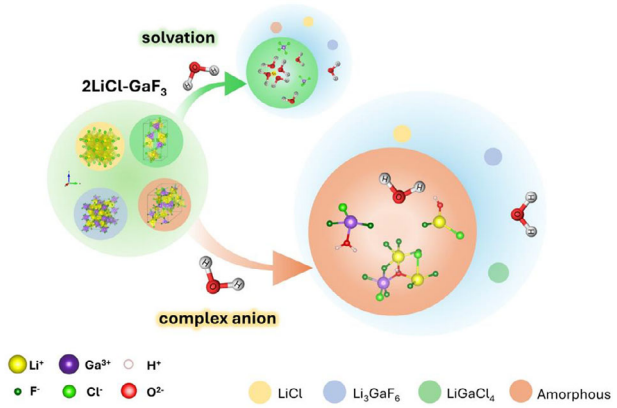

FIGURE 6 | Mechanisms of water incorporation into 2LiCl–GaF_3. Hydration induces partial solvation throughout 2LiCl–GaF_3 and predominantly drives hydroxyl–cation coordination within the dominant amorphous matrix.

that coordinate with Ga and Li. This hydroxylation reorganizes the local coordination network, weakens Li–anion interactions, and flattens the potential-energy landscape, resulting in fast Li-ion migration.

These complementary mechanisms overturn the conventional view of hygroscopicity as detrimental, instead redefining moisture as a controllable design parameter for optimizing transport and mechanical behavior. Combined experimental and AIMD analyses show that oxygen incorporation diversifies the Ga-centered coordination, enhances Li-ion diffusivity, and introduces structural compliance through amorphization. The resulting material achieves simultaneous improvements in conductivity and deformability.

This concept of water-mediated oxygen incorporation establishes a unique design principle that leverages controlled hydration to engineer mechanically compliant, fast ion-conducting oxyhalide frameworks. By transforming moisture from a limitation into a functional design tool, this strategy offers a scalable pathway toward stable, high-performance solid electrolytes and provides conceptual guidance for the development of next-generation mixed-anion solid electrolytes.

4 | Experimental Section

4.1 | Synthesis

The water-controlled 2LiCl–GaF_3 samples were prepared by planetary milling (Fritsch, Pulverisette 7PL) using LiCl (Alfa-Aesar, 99.995%) and GaF_3 (Alfa-Aesar, 99.85%) as precursors. All precursors were dried at 120°C for 12 h. Stoichiometric amounts of the precursors were inserted into an Ar-filled silicon nitride jar and mixed at 700 rpm for 10 min, followed by a 5 min rest for 18 h. After milling, the obtained 2LiCl–GaF_3 was dried at 120°C for 12 h to remove residual water. To control the water content, DI water or D_2O was mixed with the as-prepared 2LiCl–GaF_3 and stored for 24 h. $LiGaCl_4$ and Li_3GaF_6 were synthesized using the same milling procedure described above, with

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LiCl (Alfa-Aesar, 99.995%), GaF_3(Alfa-Aesar, 99.85%), and $GaCl_3$ (Alfa-Aesar, 99.99%) as precursors.

4.2 | Electrochemical Analysis

The ionic conductivity was evaluated by potentiostatic electrochemical impedance spectroscopy (PEIS) using VMP-300 (Bio-logic). The frequency range was 7 MHz to 0.1 Hz with an AC perturbation amplitude of 10 mV, measured at various temperatures (−30°C to 60°C) using a SUS | SE | SUS cell configuration.

The lithium-ion transference number (t^{Li+}) was determined using the Bruce–Vincent method. A symmetric Li | LPSCl | electrolyte | LPSCl | Li cell for powder samples (0 wt.% and 0.5 wt.% 2LiCl–GaF_3) and Li | garnet | electrolyte | garnet | Li cell for clay samples (1 wt.%, 2 wt.%, 3 wt.%, 5 wt.%, and 10 wt.% 2LiCl–GaF_3) were assembled in an Ar-filled glovebox. A DC potential difference of 0.05 V (for 0 wt.%–5 wt.% 2LiCl–GaF_3) and 1 V (for 10 wt.% 2LiCl–GaF_3) was applied across the cell using VMP-300 (Bio-Logic). The initial current (I_0) and the steady-state current (I_{ss}) were recorded, along with the interfacial resistance values before (R_0) and after (R_{ss}) DC polarization, obtained from PEIS from 7 MHz to 0.1 Hz with a perturbation amplitude of 10 mV at 25°C.

The electrochemical stability window was evaluated by cyclic voltammetry (CV) using a VMP-300 (Bio-Logic) at a scan rate of 0.1 mV s^{-1} within 0–4.8 V (vs. Li). The composite was prepared by mixing 2LiCl-GaF_3 electrolyte with 1.5 wt.% carbon nanofibers (CNF) in an agate mortar for 10 min. Cell was assembled with the configuration composite (10 mg) | Li_3InCl_6 (60 mg) | LPSCl (70 mg) | In–Li. The cell was uniaxially pressed at 300 MPa for 2 min during fabrication, and a stack pressure of 13 MPa was maintained during the measurement.

For the galvanostatic full-cell measurements, a cathode composite was prepared by mixing $LiFePO_4$ (LFP), 2LiCl-GaF_3 electrolyte, and carbon nanofibers (CNF) in a weight ratio of 50:45:5. The mixture was ground in a n agate mortar for 10 min to ensure homogeneity. An all-solid-state cell was assembled in a 13 mm polyether ether ketone (PEEK) mold with a stacking sequence of Li-In | LPSCl (70 mg) | Li_3InCl_6 (60 mg) | cathode composite (15 mg). The fabrication pressure was applied at 300 MPa for 2 min, and a uniaxial stack pressure of 70 MPa was maintained during cycling. Galvanostatic cycling was performed at 60°C using a battery cycling system (WBCS 3000, WonATech). The cell was cycled in a voltage range of 1.88–3.38 V (vs. Li+/Li-In), with current density calculated based on the mass of LFP (1C = 150 mA g^{-1}).

4.3 | Characterization

The structure of 2LiCl–GaF_3 was investigated using X-ray diffraction (XRD). XRD patterns were collected in 2θ ranges from 10° to 70° using SmartLab D/MAX2500V/PC (RIGAKU, Japan) diffractometer with Cu Kα source. Synchrotron X-ray diffraction was performed at the Pohang Accelerator Laboratory (PAL) 6D and 2D beamline. To minimize air exposure, the samples were sealed in capillaries and analyzed using a wavelength of 1.5406 Å (for 6D) and 0.7 Å (for 2D).

Ga K-edge X-ray absorption spectroscopy (XAS) was performed at the Pohang Accelerator Laboratory (PAL) 6D and 7D XAS beamline. Samples were sealed with Kapton tape to prevent air exposure. A Si (111) double-crystal monochromator was used in transmission mode. The obtained data were processed using Athena software. WT of k3 -weighted Ga K-edge EXAFS data was carried out using the Igor program.

Cl K-edge X-ray absorption spectroscopy (XAS) was performed at the 1C Tr-XRS beamline of the Pohang Accelerator Laboratory (PAL). A Si (111) double-crystal monochromator was used in fluorescence mode.

The soft X-ray absorption spectroscopy measurements were performed at the 10D beamline of the Pohang Accelerator Laboratory (PAL). The O K-edge XANES spectra were collected using TEY modes in an ultrahigh vacuum at room temperature. XANES data were treated with the Atheta software.

The microstructure of water-controlled 2LiCl–GaF_3 was examined using scanning electron microscopy (SEM). The images were acquired using SU7000(Hitachi, Japan) at an accelerating voltage of 3 kV.

The glass transition temperatures of water-controlled 2LiCl–GaF_3 were gained using differential scanning calorimetry (DSC). Samples were loaded and heated at a ramp rate of 5°C min^{-1} from −80°C to 300°C using the Q200 (TA Instruments, USA) apparatus.

Thermogravimetric analysis (TGA) was performed with a Q500 (TA Instruments, USA). Samples were loaded and heated at a ramp rate of 5°C min^{-1} from 25°C to 500°C.

X-ray photoelectron spectroscopy (XPS) measurement was performed using a K-alpha equipment (Thermofisher Scientific, USA) with a 250 mm Rowland circle monochromator. To minimize air exposure, all samples were loaded into a vacuum transfer holder in an Ar-filled glovebox.

Magic-angle spinning nuclear magnetic resonance (MAS NMR) measurements were performed on a 500 MHz Avance III HD Bruker solid-state NMR spectrometer at the National Center for Inter-University Research Facilities (NCIRF), Seoul National University. All samples were sealed inside Kel-F inserts in an Ar-filled glovebox to avoid air exposure. The MAS ^{7}Li solid-state NMR spectra were obtained at a spinning speed of 10 kHz, pulse time of 2 μs, and delay time of 5 s. The MAS ^{71}Ga solid-state NMR spectra were obtained at 8 kHz spinning with hahn-echo pulse, a pulse time of 3 μs, and a delay time of 0.1 s. The ^{2}H MAS NMR spectra were obtained at 8 kHz, pulse time of 2 μs, and delay time of 1 s.

Time-of-Flight Secondary Ion Mass Spectrometry (TOF-SIMS) measurement was performed on TOF.SIMS-5 at the National Center for Inter-University Research Facilities (NCIRF), Seoul National University. The sample was sealed in the glovebox using

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a holder. The Bi source was used as an analyzer with a beam energy of 30 KeV.

The apparent density was measured in a symmetric stainless-steel configuration (SUS | 2LiCl–GaF_3 | SUS) under uniaxial compression. A weighed amount of water-controlled 2LiCl–GaF_3 sample was loaded into a 13 mm Peek and sandwiched between stainless-steel anvils. A pressure of 100 MPa was applied and maintained for 2 min to reach thickness stabilization. The pellet thickness(t) under pressure was measured, and the apparent density was calculated by $\rho = m/(A \cdot t)$, where $A = \pi(d/2)^2$.

4.4 | Method

4.4.1 | Theoretical Calculations

First-principles calculations were conducted using the Vienna Ab initio Simulation Package (VASP) [66]. The generalized gradient approximation (GGA) exchange correlation with the Perdew–Burke–Ernzerhof (PBE) functional [67] was adopted, along with the projector-augmented wave (PAW) method. A plane-wave cut-off energy of 520 eV was used, with the cell shape, cell volume, and atomic positions of each structure fully relaxed until the forces on each atom were below 0.05 eV/Å. The initial structure of Li_3GaF_6 and $LiGaCl_4$ was obtained from the Materials Project database (mp-15558 and mp-28341), and a supercell (lattice parameter > 10 Å) was used for all DFT simulations. For crystalline Li_3GaF_6 and $LiGaCl_4$, we employed the enumeration method to systematically explore possible atomic arrangements of oxygen (with hydrogen) and additional lithium by considering different combinations of atoms at partially occupied sites. The Python Materials Genomics (Pymatgen) [68] package was used for efficient enumeration. To identify additional Li sites within the structures, the TopographyAnalyzer module from Pymatgen was utilized to preselect topologically feasible sites for Li insertion. Initially, structure files for the 30 configurations with the lowest Ewald summation energies, based on atomic positions, were extracted. These structures were then subjected to structural relaxation using identical DFT parameters, and the configuration with the lowest total energy was selected for further analysis.

For amorphization, melt-quenching Ab initio molecular dynamics (AIMD) simulations are conducted. These simulations used an NVT ensemble with a Nose–Hoover thermostat having a period of 80 fs [69]. Crystalline $LiGaCl_4$ and Li_3GaF_6 were subjected to heat-up simulations, incrementally increasing the temperature from 100 K to 3000 K during 2 ps. Following the heat-up process, diffusion simulations were conducted at 3000 K for a duration of 10 ps. Subsequently, quenching simulations were performed, reducing the temperature from 3000 K to 100 K over a time span of 2 ps at every 10 ps interval of the diffusion process. The quenched structures obtained from each cycle were fully relaxed using density DFT with consistent computational parameters. AIMD simulations for calculating the Li^+ diffusivity were performed to reveal the conduction mechanism. The crystalline and amorphous structures have lattice parameters larger than 10 Å in each direction, and a Γ-centered 1 × 1 × 1 k-point grid was used. The heat-up process was executed for each supercell by increasing the temperature from 100 K to 600 K over 2 ps. AIMD simulations were conducted with a 2-fs interval time step for 200 ps at different holding temperatures, and diffusivities (D) were determined by linear fitting of the mean square displacement (MSD) of lithium ions using the following equations:

$$MSD = \frac{1}{N}\sum_{i=1}^{N}\left|r_i(t+\Delta t) - r_i(t)\right|^2 \tag{1}$$

$$D = \frac{1}{2dtN}\sum_{i=1}^{N}\left|r_i(t+\Delta t) - r_i(t)\right|^2 \tag{2}$$

where r_i is the position of the ith ion at time t, Δt is the time step, N is the number of Li atoms in the supercell structure, and d is the dimensionality factor. The MSD and diffusivities were analyzed using a diffusion analyzer module [70] in Pymatgen. Diffusion channel size analysis based on the Li-ion trajectories generated from AIMD simulations was performed using the topological analysis package Zeo^{++} [71].

The coordination number (N_{A-B}) between atomic species A and B was calculated by integrating the partial radial distribution function $g_{A-B}(r)$ up to the first minimum, as expressed by:

$$N_{A-B} = 4\pi\rho_B \int_0^{r_{min}} r^2 g_{A-B}(r)\,dr \tag{3}$$

where ρ_B is the number density of species B, and $r_{\min}$ represents the first minimum following the primary peak in $g_{A-B}(r)$. Here, the inclusion of the r^2 term reflects the spherical volume element and makes the coordination number inherently sensitive to the spatial distribution of neighboring atoms within the first coordination shell. The coordination fraction of each anion type around Li^+ was further quantified by normalizing the corresponding coordination numbers, allowing statistical evaluation of the relative Li–anion coordination ratios in each simulated structure. Accordingly, the coordination fractions f_{A-B} represent relative, r^2-weighted contributions of different anion species within the Li^+ first coordination shell, rather than absolute coordination numbers. This normalization enables a meaningful comparison of anionic coordination environments that is decoupled from variations in absolute coordination magnitude and Li-ion mobility.

$$f_{A-B} = \frac{N_{A-B}}{\sum_i N_{A-i}} \tag{4}$$

Author Contributions

Y.K. and S.-K.J. conceived and supervised the study, designed the research, and wrote the manuscript. J.-S.K. and D.-H.S. performed the DFT calculations and AIMD simulations and contributed to manuscript preparation. K.H.S. measured the Li^+ transference number. J.C. conducted SEM characterization and analysis. B.J.P. and C.P. assisted with electrochemical characterization. J.L. provided guidance on the solid-state NMR experiments and their interpretation. All authors discussed the results and approved the final manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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